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EUROPEAN PATENT OFFICE
MUNICH 80298
GERMANY

Online filing

European patent application (Euro/PCT) No. 22820344.4 of June 10, 2022
In the name of : THE KITASATO INSTITUTE
Title : KIT FOR DIAGNOSIS OF CANCER AND USE THEREOF
Our ref. : B79149EP D44435

Dear Madam, dear Sir,

We refer to your communication pursuant to Rules 70(2) and 70a(2) EPC issued for the above-mentioned patent application.

The Applicant hereby declares that they maintain and wish to proceed further with the above-referenced patent application. The EPO is reminded that the designation/examination fee was paid upon filing of the referenced application.

Moreover, in response to this communication, the Applicant files the enclosed amended set of claims along with the following observations.

Amendments

The following amendments have been made:

- Claim 1 has been amended to include the subject matter of previous claim 2;
- Claim 3 (new claim 2) has been amended to include the subject matter of previous claim 4;
- Claim 5 (new claim 3) has been amended to include the subject matter of previous claim 6;
- Claim 7 (new claim 4) has been amended to include the subject matter of previous claim 8. In addition, new claim 4 has been redrafted as use claim;
- Claim 9 (new claim 5) has been amended to include the subject matter of previous claim 10. In addition, new claim 5 has been redrafted as use claim;
- Claim 11 (new claim 6) has been amended to include the subject matter of previous claim 12, and
- Previous claims 2, 4, 6, 8, 10, and 12 have been cancelled.

These amendments comply with the requirement of Article 123(2) EPC.

Any amendment to the claims is not to be construed as abandonment of any subject matter, either for the purposes of this application or that of a divisional application.

Observations

The pending claims are objected to for lack of novelty and inventive step.

- ***D1: Santa Cruz: "SSB-2 (32-K)***

The Examiner stated that **D1** discloses a kit comprising anti-human SPSB2-antibodies suitable for detecting SPSB2 in pancreatic cancer samples and thus anticipates the subject matter of claims 7-10.

The Applicant points out that that pancreatic cancer samples are not specifically described in **D1**.

Nonetheless, in order to move forward with the examination, previous claims 7 to 10 (new claims 4 and 5) have been amended as use claims and the kit is no longer claimed as a product *per se*.

In addition, these claims have been limited to specify that the cancer is bladder cancer or hepatocellular carcinoma. Pancreatic cancer is no longer recited in the claims.

Therefore, the claims enclosed herewith are novel over **D1**.

- ***D2: US 2013/196876 A1***

Regarding **D2**, the Examiner indicates that this document discloses that SPSB2 as well as autoantibodies to SPSB2 were found as differentially expressed between pancreatic carcinoma patients and controls and that it also discloses staging, treatment and prognosis. **D2** is thus considered to anticipate the subject matter of claims 1-12.

As indicated above, pancreatic cancer has been deleted from the claims. Therefore, the amended claims are novel over **D2**.

- ***D3: ZENG CHENJIE ET AL.***

The Examiner considers that **D3** anticipates the subject matter of claims 1, 3, 5 and 7-10 because this document discloses kits comprising anti-SPSB2- antibodies, as well as probes suitable for detecting SNPs in colorectal carcinoma patients.

The Applicant points out that the cancer in the amended claims has been restricted to cancer other than colorectal carcinoma (bladder cancer or hepatocellular carcinoma). Therefore, the amended claims are novel over **D3**.

- ***D4: PING JIE ET AL.***

The Examiner indicates that **D4** discloses methods involving gene expression analyses in breast cancer samples wherein RNA-Seq of total RNA isolated from formalin-fixed paraffin-embedded (FFPE) breast cancer tissues is investigated and wherein SPSB2 is significantly differentially expressed in said samples. **D4** is thus considered as anticipating the subject matter of claims 1 and 11.

Because the claims have been restricted to cancer other than breast cancer (bladder cancer or hepatocellular carcinoma), they are novel over **D4**.

From the above, it appears that the amended claims enclosed herewith are novel over **D1-D4** and meet the requirement of Article 54 EPC.

2. In addition, the Applicant submits that none of documents **D1-D4** suggests providing a method for collecting data for determining whether or not a subject is affected by cancer or method for predicting a prognosis of a patient with cancer or method for screening an anti-cancer agent or the use of kit for these purposes, wherein the cancer is bladder cancer or hepatocellular carcinoma and the method involves measuring an expression level of an SPSB2 protein or an SPSB2 gene.

Therefore, the skilled artisan who has read documents **D1-D4** cannot arrive at the present invention as amended and the present invention thus involves an inventive step.

Consequently, the amended claims enclosed herewith also meet the requirements of Article 56 EPC.

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In light of these submissions, the application is considered to be in compliance with the provisions of the EPC.

In the event that the Examining Division intends to refuse the application without sending any further communication, we request oral proceedings by videoconference.

Yours sincerely,

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Encl.:
Amended claims (clean and marked versions)